1. JKSST xedni analysis: skeptical Lean audit

Kernel-checked formal audit, 2026-11-01

1.1. Result

The Lean suite proves finite packing, cyclic extraction, exact event counts, and formal replacements for probabilistic and runtime steps needed to make the JKSST xedni analysis precise. It also gives counterexamples to several overbroad readings. The paper's final conclusion remains conditional on a family-uniform selected-lift height/discriminant comparison, Lang or a direct height gap, good reduction and subgroup correspondence, useful-relation selection, a bounded cofactor, and a repeated-trial hazard theorem.

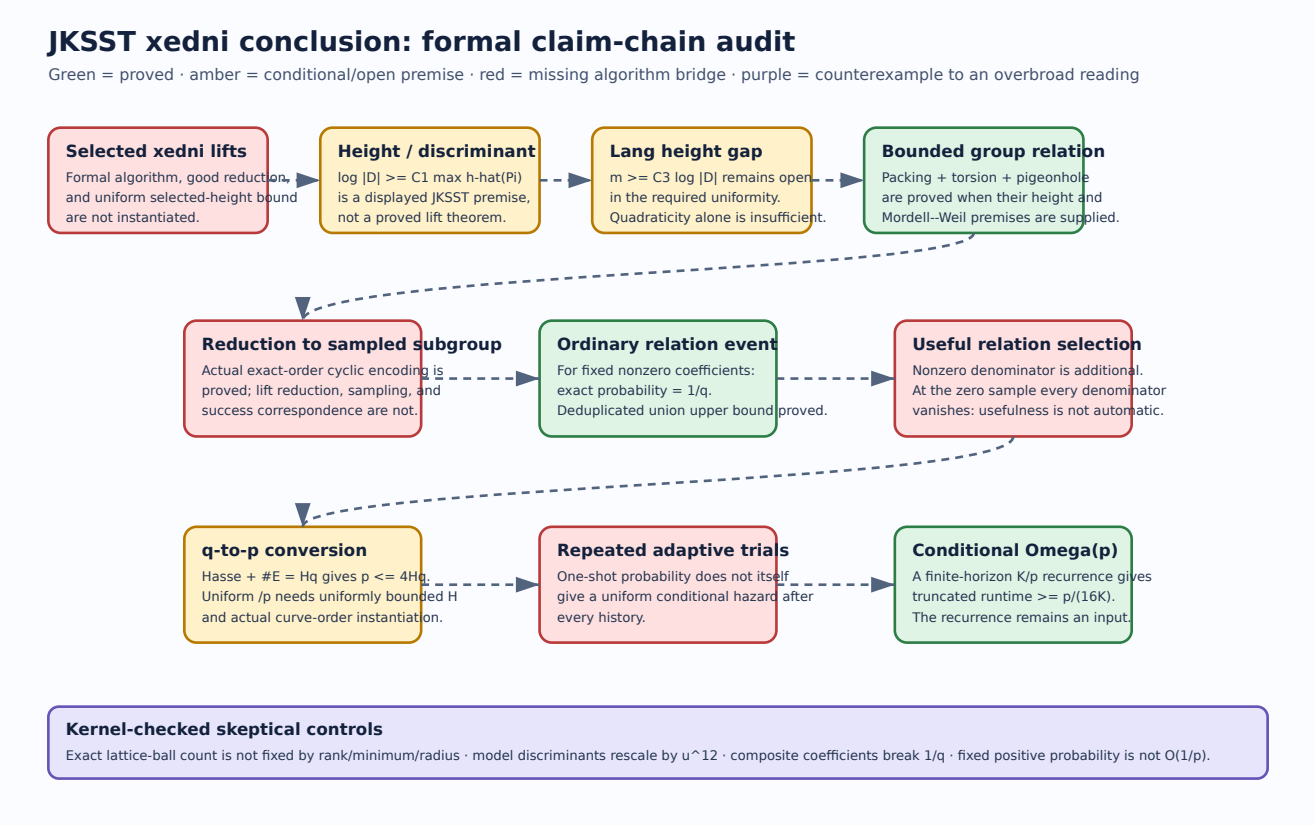


Figure 1: Claim-chain status. Green boxes are formal implications; amber boxes are explicit arithmetic premises; red boxes are missing algorithm interfaces; purple records kernel-checked counterexamples.

1.2. Source correspondence

JKSST Theorem 4.1 starts "under certain plausible assumptions." Lemma 4.1 assumes `log |D| >= C1 max h-hat(Pi)` and Lang's conjecture. Its finite-field argument calls `(2 C2)^r / p` an approximate probability. The audit replaces that step with exact finite-field cardinalities and keeps subgroup order `q` distinct from base-field prime `p`.

The actual canonical-height construction is tied to Mathlib's rational Weierstrass point group. Its all-multiples counterexample does not claim that xedni's selected lifts contain those multiples. The formal result therefore corrects an unconditional reading without asserting that xedni succeeds.

1.3. Decisive results

- Exact coefficient cube: `(2 C + 1)^r`, with the zero vector removed.
- Same rank, minimum, and radius can give different exact lattice-ball counts.
- Finite repair: `#S <= T (1 + 2R/delta)^k` with every hypothesis visible.
- Fixed ordinary relation probability: `1/q`; fixed useful relation: `(q-1)/q^2`.
- A bounded relation does not automatically select a useful denominator.
- Actual exact-order curve-point encoding supports correct log extraction.
- Hasse plus `#E=Hq` yields a conditional `p <= 4Hq`, not `p approximately q`.
- A uniform finite-horizon hazard recurrence yields a literal truncated `Omega(p)` lower bound; a one-shot event bound is not that recurrence.

1.4. Trust and limits

The project builds under Lean/Mathlib 4.34.1. The axiom ledger reports only `propext`, `Classical.choice`, and `Quot.sound`; there are no project axioms, `sorry`, `admit`, or `native_decide` proofs. Passing this gate certifies the formal declarations, not the still-missing correspondence to the lift selection and adaptive algorithm.

Primary source: M. J. Jacobson, N. Koblitz, J. H. Silverman, A. Stein, and E. Teske, “Analysis of the Xedni Calculus Attack,” DCC 20 (2000), 41–64, DOI 10.1023/A:1008312401197. Full source and theorem-level mapping are in `README.md` and `PROOF_MAP.md` beside this report.

1.5. Open end-to-end obligations

- Specify the lift-selection algorithm and prove a family-uniform comparison with a global minimal discriminant.
- Prove Lang for that family, or replace it with an unconditional direct canonical-height gap.
- Instantiate finite generation, rank, separation, torsion fibers, and the exact finite packing hypotheses for the selected lifts.
- Connect good reduction to the sampled prime-order subgroup and prove the sampling correspondence.
- Show every algorithmic success selects a bounded scalar relation and, for the tighter event, a nonzero denominator.
- Prove a uniform cofactor bound before replacing subgroup order `q` by base-field prime `p`.
- Derive the history-uniform conditional-hazard recurrence for the adaptive algorithm; a one-shot relation rate does not supply it.
- If the cubic dimension step is used, prove full row rank for the actual ten evaluation conditions.